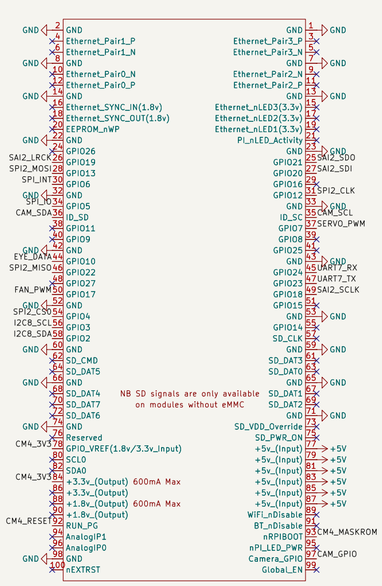
Radxa CM4 on DF40C sockets

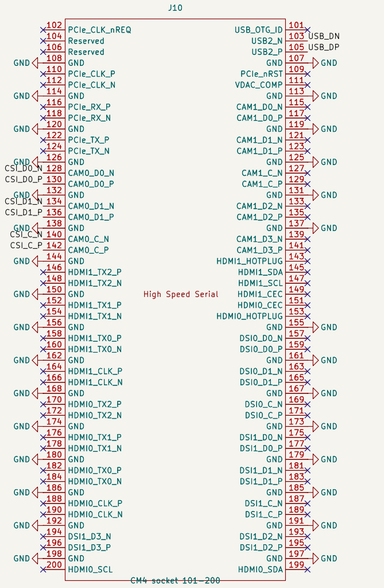
Pin allocation: docs/head-pcb.md section 4. Unused pins not connected, as on the NANO-A

J9



CM4 socket 1–100

J10



CM4 socket 101–200

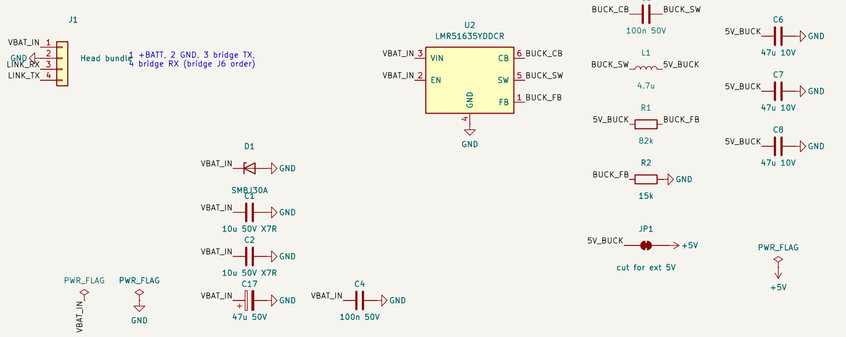
GPIO_VREF (78) on the module's own 3.3 V output (84; 86 is the same rail, left open): 3.3 V GPIO, as the NANO-A's H2 jumper

H5

Radxa CM4

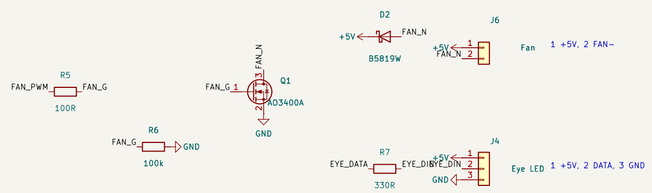
Head bundle in, 5 V buck

LMR51635Y, 1.1 MHz, 4.7 uH (Ti table 8-2: 3.3 uH), 82k/15k sets 5.17 V, EN tied to VIN. F6 (2 A) on the bridge fuses this feed.
JP1 joins the buck to the 5 V rail: cut it to feed 5 V from outside (J12). The feedback divider stays on the buck side



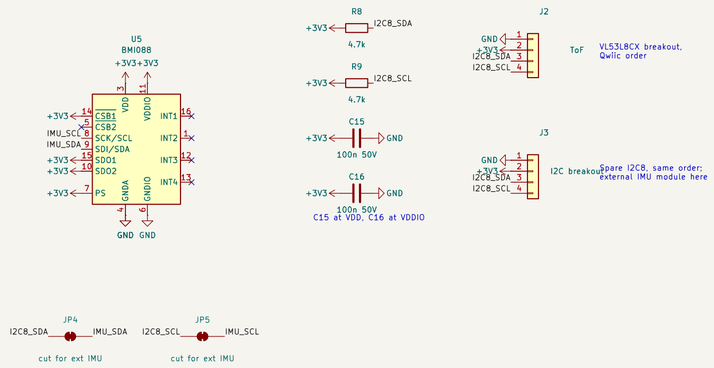
Fan and eye LED

Fan: low side, PWM1_CH2 (pin 50). Eye: WS2812-type, SPI1 MOSI (pin 44), 3.3 V data



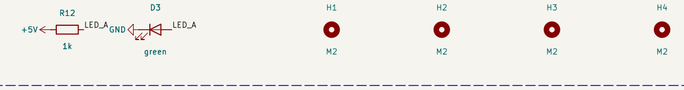
I2C8: head IMU, ToF, breakout

BMI088 on the board. I2C: PS and CSB1 high, CSB2 open (allowed in I2C mode): SDO1 and SDO2 high: accelerometer 0x19, gyroscope 0x69 (strapped so the pin rows need no vias between alternate pins).
R8, R9: the bus pull-ups (the ToF breakout adds 10k). Cut JP4 and JP5 to use an external IMU module on J3



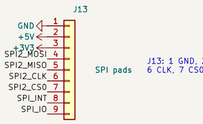
Power LED, mounting holes

M2 holes, one per corner, not connected



SPI breakout

SPI2 M1 for a future SPI device, untested. Wire pads on the bottom, under the module, along the top edge.
CLK GPIO12 (CM4 31), MOSI GPIO13 (28), MISO GPIO22 (46), CS0 GPIO4 (54), INT GPIO6 (30), spare GPIO5 (34)



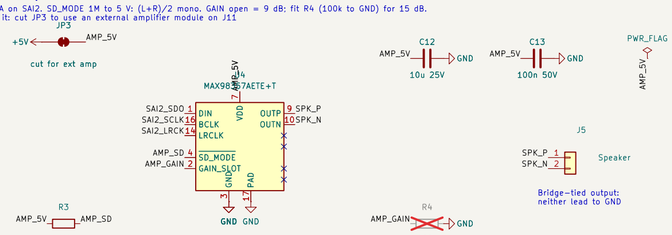
3.3 V

AMS1117-3.3 from 5 V, as on the NANO-A. Camera, ToF, IMU, mic, I2C breakout



Speaker amplifier

MAX98357A on SAI2. SD_MODE 1M to 5 V: (L+R)/2 mono. GAIN open = 9 dB: fit R4 (100k to GND) for 15 dB.
JP3 feeds it: cut JP3 to use an external amplifier module on J11

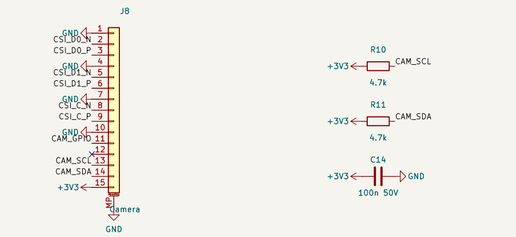


Microphone

INMP441 on SAI2. L/R tied to GND on the mic side (left slot)

Camera FPC

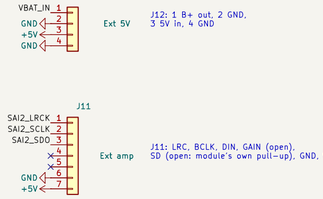
Pi 15-pin camera order. CAM0 lanes (Radxa CSI1), I2C6 with 4.7k pull-ups, CAM_GPIO enable. 100 ohm pairs



Pin 12 (camera clock) not connected: the module has its own oscillator

External fallbacks

Wire pads, 2.54 mm. J12: cut JP1 and feed 5 V from an external step-down. J11: cut JP3 and wire an amplifier module, pins in the generic module's order. J3 (section 6) takes an external IMU module



Link series resistors, test pads

330R in each UART7 line limits current into an unpowered CM4 from the bridge. TP1/TP2 (bottom): short to TP3 to reset; hold TP2 low through a reset for maskrom. TP4–TP6 (bottom): USB D−, D+, GND of the download port, for rkdeveloptool maskrom. TP7: GPIO7 / PWM0_CH0_M1 (pin 37) for a servo

Bootstrapped by scripts/kicad/gen_head_sch.py: edit this file from now on
Design note: docs/head-pcb.md
File: head.kicad_sch
Title: Dukki head carrier
Size: A2 | Date: 16/10/2026 | **Rev: B**
KiCad E.D.A. 10.0.0 | Id: 1/1